2	**(a)**	**(i)**	*k* is the reciprocal of the gradient of the graph	C1	
			$k = \{32 / (4 \times 10^{-2}) = \}\ 800\ \text{N m}^{-1}$	A1	[2]
		(ii)	*either* energy = average force × extension *or* $\tfrac{1}{2}kx^2$		
			or area under graph line	C1	
			energy $= \tfrac{1}{2} \times 800 \times (3.5 \times 10^{-2})^2$ *or* $\tfrac{1}{2} \times 28 \times 3.5 \times 10^{-2}$	M1	
			energy = 0.49 J	A0	[2]
	(b)	**(i)**	momentum before cutting thread = momentum after	C1	
			$0 = 2400 \times V - 800 \times v$	M1	
			$v / V = 3.0$	A0	[2]
		(ii)	energy stored in spring = kinetic energy of trolleys	C1	
			$0.49 = \tfrac{1}{2} \times 2.4 \times (\tfrac{1}{3} v)^2 + \tfrac{1}{2} \times 0.8 \times v^2$	C1	
			$v = 0.96\ \text{m s}^{-1}$	A1	[3]
			(if only one trolley considered, or masses combined, allow max 1 mark)		

3 **(a)** **(i)** $v^2 = 2as$